Why Imaginary Numbers Matter in Quantum Mechanics

Syed Hussain Haider

3rd Year Physics with Theoretical Physics (BSc)

King's College London

k23102240@kcl.ac.uk

Where Do Imaginary Numbers Come From?

- Squares of real numbers are always non-negative:

$$2^2 = 4, \qquad (-2)^2 = 4$$

- In general:

$$x^2 \geq 0 \quad \text{for all } x \in \mathbb{R}$$

- Taking square roots:

$$\sqrt{x^2} = |x|$$

What About Negative Numbers?

- If $x^2 \geq 0$ for every real x, then:

$$\sqrt{x^2} = |x|$$

- But then what is:

$$\sqrt{-1}\,?$$

- No real number has a square equal to a negative value:

$$|y| = -1 \quad \text{has no real solution}$$

- So we define a new number:

$$i \equiv \sqrt{-1}$$

Why i Is Unavoidable

- When you follow the rules of algebra consistently, expressions like

$$\sqrt{-1}$$

  appear naturally.
- This happens even in problems that start with only real numbers.
- To handle this cleanly, mathematicians introduced a new number i.
- Without i, we are forced to create many separate rules (as we'll see next).

Before Complex Numbers: Quadratics Were a Mess

- In the 1500s, negative numbers were not accepted.
- So mathematicians solved quadratics using **six separate cases**:

$$
\begin{aligned}
&(1) \quad x^2 + bx = c \\
&(2) \quad x^2 = bx + c \\
&(3) \quad x^2 + c = bx \\
&(4) \quad x^2 - bx = c \\
&(5) \quad x^2 - c = bx \\
&(6) \quad x^2 = c - bx
\end{aligned}
$$

- Each form required its own method of solution.
- All because mathematicians wanted to avoid negative numbers and $\sqrt{-1}$.

After Complex Numbers: One Formula Solves Everything

- Once negative numbers and i were accepted, all six cases collapsed into a single universal formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

- The discriminant $b^2 - 4ac$ can be positive, zero, or negative.
- Complex numbers handle all possibilities seamlessly.
- The messy case-by-case approach disappears.

Are Imaginary Numbers Physical?

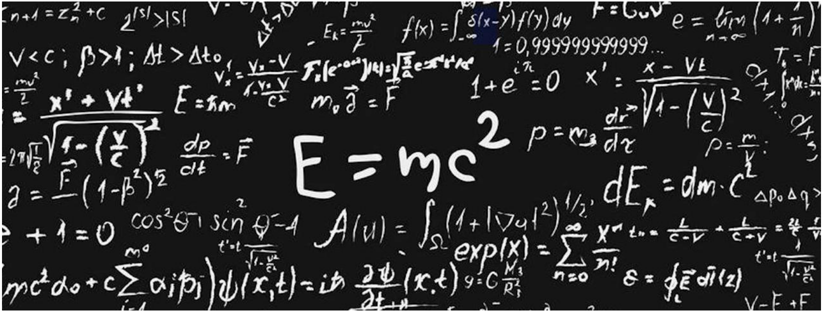
- Do imaginary numbers describe anything real?
- Are they just a mathematical trick?
- Were they “invented” only to save equations?
- Or do they reflect something deeper about nature?

Are They Really Just “Imaginary”?

- We often hear that imaginary numbers are not “real.”
- But physics uses them in its most fundamental equations.
- We will see that i is not a trick:
- **the universe behaves the way it does because of i.**

Why Differential Equations?

- Physics is written in differential equations.
- Many of these equations naturally produce i.
- Changing a real coefficient to an imaginary one transforms decay into oscillation.



A Tale of Two Equations

$$\frac{\partial\psi}{\partial t} = \frac{i\hbar}{2m}\frac{\partial^2\psi}{\partial x^2}$$

Schrödinger's equation

(with $V(x) = 0$)

$$\frac{\partial u}{\partial t} = D\,\frac{\partial^2 u}{\partial x^2}$$

Heat equation

The Key Identification

- Compare the two equations carefully.
- They differ only in the constant multiplying the second derivative.
- If we choose

$$D = \frac{i\hbar}{2m},$$

then the heat equation and Schrödinger's equation become the same equation.

From Equations to Solutions

- Since the two equations have the same form,

$$\frac{\partial}{\partial t}(\,\cdot\,) = D\,\frac{\partial^2}{\partial x^2}(\,\cdot\,),$$

  their solutions will also have the same structure.

- So we first solve the heat equation (real D).
- Then we simply replace

$$D \;\longrightarrow\; \frac{i\hbar}{2m}$$

  to obtain the solution for Schrödinger's equation.

Oscillation vs Decay: The Role of i

- **Heat equation mode:**

$$e^{-Dk^2 t} \quad \Longrightarrow \quad \text{decays exponentially}$$

- **Schrödinger equation mode:**

$$e^{-i\frac{\hbar}{2m}k^2 t} = \cos\left(\frac{\hbar k^2}{2m}t\right) - i\sin\left(\frac{\hbar k^2}{2m}t\right)$$

oscillatory, pure phase rotation

- **Key insight:**

$$e^{-Dk^2 t} \quad \longrightarrow \quad e^{-iD_{\text{real}}k^2 t}$$

Just replacing D by iD transforms decay into oscillation.

Solutions of the Two Equations

- **Schrödinger's equation solution:**

$$\psi(t) \;\propto\; e^{-\,i\,\frac{\hbar}{2m}\,k^2 t}$$

Oscillatory, pure phase rotation.

- **Heat equation solution:**

$$u(t) \;\propto\; e^{-\,D\,k^2 t}$$

Decaying, modes die exponentially.

General Form of Schrödinger Solutions

- Solutions of the time–dependent Schrödinger equation can always be written as:

$$\psi(t) = \sum_n A_n(x)\, e^{-iE_n t}$$

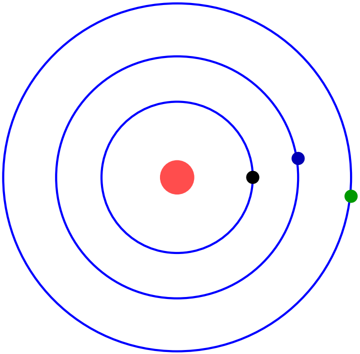
- Each term has two parts:
  - $A_n(x)$: the spatial shape of the n-th mode
  - $e^{-iE_n t}$: oscillatory phase determined by the energy
- This means any quantum state is a **combination of allowed energy states**.
- These allowed energies are not continuousthey come in discrete levels.

Discrete Energy Levels

- In quantum mechanics, a particle cannot have arbitrary energies.
- Only certain values are allowed:

$$E_0 < E_1 < E_2 < \cdots$$

- Each allowed energy corresponds to an eigenstate ϕ_n.
- Example: electrons in an atom occupy discrete levels.
- These levels form the "basis directions" from which all states are built.



- ● E_1 (1st excited level)
- ● E_2 (2nd excited level)
- ● E_3 (3rd excited level)

Eigenstates as Directions

- Think of the eigenstates ϕ_n as "directions" in a (very large) vector space.
- Like the 3D basis:

$$\hat{x},\ \hat{y},\ \hat{z}$$

but in quantum mechanics there are many more "axes":

$$\phi_0,\ \phi_1,\ \phi_2,\ \ldots$$

- Any wavefunction can be written as a vector in this space:

$$\psi = c_0\,\phi_0 + c_1\,\phi_1 + c_2\,\phi_2 + \cdots$$

- Understanding this picture makes the subtraction method (Gram–Schmidt) intuitive.

Why Schrödinger Evolution Isn't Enough

- Many quantum systems do not have analytic wavefunctions.
- To find ground and excited states, we often rely on numerical methods.
- But real-time Schrödinger evolution is oscillatory:

$$\psi(t) = \sum_n A_n(x)\, e^{-iE_n t}$$

- Which we can rewrite as:

$$e^{-iE_n t} \;=\; \cos(E_n t) - i \sin(E_n t)$$

- All components oscillate forever, nothing decays or separates.

Oscillation Does Not Isolate States

- Real-time evolution preserves **all** energy components.
- No mode decays, no mode dominates.
- This makes it difficult to extract a specific state (e.g. the ground state).
- We need an evolution that **separates** modes rather than preserving them.

A Striking Similarity

- We saw that Schrödinger's equation is almost identical to the heat equation.
- The heat equation has solutions of the form:

$$e^{-Dk^2 t} \quad \text{(exponential decay)}$$

- Decay naturally removes high-energy components.
- Oscillation, in contrast, keeps everything alive.

Why Decay Helps

- Schrödinger evolution:

$$e^{-iE_n t} \quad \text{oscillatory, no suppression}$$

- Heat evolution:

$$e^{-At} \quad \text{decaying, higher } A \text{ dies faster}$$

- Diffusion introduces a clear direction: low-energy modes survive.
- **This leads to a natural question:**
- **Can we make Schrödinger behave like a heat equation?**

Wick Rotation: Imaginary Time

- Replace real time by imaginary time:

$$it \;\to\; \tau$$

- Then by the chain rule:

$$\frac{\partial}{\partial t} = -i\,\frac{\partial}{\partial \tau}$$

- Under this substitution, Schrödinger's equation becomes

$$\frac{\partial \psi}{\partial \tau} = -\frac{\hbar}{2m}\,\frac{\partial^2 \psi}{\partial x^2}$$

- This is precisely a **diffusion (heat) equation**.

What Wick Rotation Does to the Solutions

- Before (real time):

$$\psi(t) = \sum_n A_n(x)\, e^{-iE_n t}$$

- After (imaginary time):

$$\psi(t) = \sum_n A_n(x)\, e^{-E_n \tau}$$

- Oscillatory phases turn into exponential decays.
- This diffusive behaviour opens the door to state extraction.

Factoring Out the Ground-State Contribution

- Start with the imaginary-time expansion:

$$\psi(\tau) = \sum_n A_n(x)\, e^{-E_n \tau}$$

- Rewrite each term by adding and subtracting E_0:

$$e^{-E_n \tau} = e^{-E_0 \tau}\, e^{-(E_n - E_0)\tau}$$

- Apply this to every term:

$$\psi(\tau) = A_0\, e^{-E_0 \tau} + A_1\, e^{-E_0 \tau} e^{-(E_1 - E_0)\tau} + A_2\, e^{-E_0 \tau} e^{-(E_2 - E_0)\tau} + \dots$$

- Now factor out the common $e^{-E_0 \tau}$:

$$\psi(\tau) = e^{-E_0 \tau} \left[A_0 + A_1\, e^{-(E_1 - E_0)\tau} + A_2\, e^{-(E_2 - E_0)\tau} + \dots \right]$$

Why Only the Ground State Survives

- All excited energies lie above the ground state:

$$E_1 - E_0 > 0, \qquad E_2 - E_0 > 0, \quad \dots$$

- Therefore each excited term carries a decaying factor:

$$e^{-(E_n - E_0)\tau} \;\longrightarrow\; 0 \qquad (\tau \to \infty)$$

- The bracket collapses to a constant:

$$\left[\, A_0 + 0 + 0 + \dots \,\right]$$

- In the large–τ limit:

$$\psi(\tau) \;\longrightarrow\; e^{-E_0 \tau}\,[A_0(x)]$$

What About Excited States?

- So far, imaginary-time evolution isolates the **ground state**:

$$\psi(\tau) \;\longrightarrow\; e^{-E_0\tau}\, A_0(x)$$

- But quantum systems have many eigenstates:

$$A_0(x),\; A_1(x),\; A_2(x),\; \ldots$$

- Can we use the same idea to extract **excited** states?
- Yes, by first removing the ground-state component:

$$\tilde{\psi}(\tau) = \psi(\tau) - \langle A_0|\psi(\tau)\rangle\, A_0(x)$$

Removing a State: Vector Analogy

- Suppose a vector in 3D is:

$$(1,\ 1,\ 1)$$

- If we want to remove the "x–direction" component, subtract:

$$(1,\ 0,\ 0)$$

- Result:

$$(0,\ 1,\ 1)$$

which has no projection on the x–axis.

- The same idea applies in Hilbert space:

$$\tilde{\psi}(\tau) = \psi(\tau) - \langle A_0 | \psi(\tau) \rangle \, A_0(x)$$

- This new wavefunction has **no component of the ground state**.
- Now imaginary–time evolution isolates the next state.

Extracting the First Excited State

- After removing the ground state:

$$\tilde{\psi}(\tau) = A_1\, e^{-E_1\tau} + A_2\, e^{-E_2\tau} + \dots$$

- Now repeat the same trick as before:

$$e^{-E_n\tau} = e^{-E_1\tau}\, e^{-(E_n-E_1)\tau}$$

- Factor out the $e^{-E_1\tau}$ term:

$$\tilde{\psi}(\tau) = e^{-E_1\tau}\left[A_1 + A_2\, e^{-(E_2-E_1)\tau} + \dots\right]$$

- As $\tau \to \infty$, only the first excited state survives:

$$\tilde{\psi}(\tau) \;\longrightarrow\; e^{-E_1\tau}\,[A_1]$$

Higher Excited States

- To obtain the second excited state, remove the first two:

$$\psi^{(2)}(\tau) = \psi(\tau) - \langle A_0|\psi(\tau)\rangle A_0 - \langle A_1|\psi(\tau)\rangle A_1$$

- Imaginary-time evolution then isolates the next state:

$$\psi^{(2)}(\tau) \longrightarrow e^{-E_2\tau} A_2(x)$$

- Continue this process:

$$A_0,\ A_1,\ A_2,\ \ldots$$

  every state can be extracted.
- Higher states require more subtractions and more computational precision.

From Intuition to Reality

- Earlier, we wrote the wavefunction in the ideal form:

$$\psi(t) = \sum_n A_n(x)\, e^{-iE_n t}$$

- In such a case, extracting states looks easy:
    - set $n = 0$ for the ground state
    - set $n = 1$ for the first excited state
    - and so on
- But this relies on knowing all $A_n(x)$ and E_n.
- **In reality, most quantum systems do not have analytic solutions.**

Why We Need Computation

- The expansions we used earlier were only to build intuition.
- In real quantum systems, we do **not** know the eigenstates $\phi_n(x)$ or energies E_n.
- Most potentials do not have closed-form solutions.
- So to apply imaginary-time evolution in practice, we must work **numerically**.

Imaginary-Time Evolution With the Hamiltonian

- We write the Schrödinger equation in imaginary time as:

$$\frac{\partial \psi}{\partial \tau} = -H\psi$$

- Here H is the Hamiltonian:

$$H = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)$$

- To solve this numerically, we:
  - discretise space on a grid,
  - approximate derivatives with finite differences,
  - and evolve ψ step by step in imaginary time.

How We Evolve the Wavefunction

- At each small imaginary-time step $\Delta\tau$, we update the wavefunction.
- We use a stable time-stepping method (Crank–Nicolson) to do this reliably.
- After every update:
  - we **normalise** ψ,
  - and compute its energy.
- Over many steps, higher-energy components decay faster than lower ones.
- The wavefunction gradually **relaxes** toward lower-energy states.

Extracting Excited States (Numerical Version)

- Once the ground state ϕ_0 has converged, we store it.
- To find the next state, we remove the ground-state component:

$$\psi_{\text{residual}} = \psi - \langle \phi_0 | \psi \rangle \phi_0$$

- We then evolve this residual again in imaginary time.
- The same decay mechanism now isolates the first excited state.
- Repeating this procedure successively yields $\phi_1, \ \phi_2, \ldots$.

Different Potentials, Same Method

- Our imaginary–time equation is:

$$\frac{\partial \psi}{\partial \tau} = -H\psi$$

where the Hamiltonian is

$$H = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} \ + \ V(x).$$

- Once the algorithm is set up, the **only part that changes** from one quantum system to another is the potential $V(x)$.
- For each new system:
  - insert the appropriate $V(x)$ into the Hamiltonian,
  - discretise it on the same spatial grid (or a modified one, if needed),
  - evolve in imaginary time using the **same procedure**.
- The kinetic term is always the same; **all the physics enters through the choice of** $V(x)$.

Quartic Potential

- One of the standard test cases in quantum mechanics:

$$V(x) = x^4 + 0.1x^2$$

- No analytic solutions — ideal for numerical methods.
- We apply imaginary-time evolution to extract the first few eigenstates.
- If our method works, the energies should match known reference values.

Quartic Potential: Results

- The potential was:

$$V(x) = x^4 + 0.1x^2$$

n	Computed E_n	Ref. E_n
0	1.065238	1.065286
1	3.306602	3.306872
2	5.747178	5.747959
3	8.351025	8.352678
4	11.095647	11.098600

The computed energies agree with reference values to ~ 4 significant figures.

Position-Dependent Mass System (PDMSE)

- Next, we applied the same imaginary-time method to a system with position-dependent mass.
- The potential was:

$$V(z) = \frac{1}{2} + \frac{3}{4}\tan^2 z + \sin z, \qquad z \in [-\pi/2,\, \pi/2].$$

n	Computed E_n	Ref. E_n
0	1.950409	1.950334
1	6.011908	6.011578
2	12.009202	12.008326
3	20.007291	20.005520
4	30.006896	30.003847

Agreement at the level of ~ 4 significant figures.

Hydrogen Atom: Soft Coulomb Potential

- Finally, we tested the method on the hydrogen atom using a softened Coulomb potential:

$$V(r) = -\frac{1}{r}.$$

- Imaginary-time evolution reproduces the expected Rydberg scaling of energies.

n	Computed E_n	Exact E_n
0	-0.497461	-0.500000
1	-0.123492	-0.125000
2	-0.057041	-0.055556
3	-0.030474	-0.031250
4	-0.021391	-0.020000

Captures the expected $-\frac{1}{2n^2}$ energy trend accurately.

From Imaginary Time to Imaginary Phase

- So far, we used imaginary time to turn oscillations into decay.
- This helped us compute:
  - ground states,
  - excited states,
  - eigenvalues of various potentials.
- But oscillations themselves are not just mathematical noise.
- **They are the entire reason quantum mechanics behaves differently from classical physics.**
- To see the physical role of complex phase, we now switch to a real experiment: the double–slit.

Classical Intuition: One Slit

- Think of light as particles (photons) hitting a screen.
- With only one slit open:

$$\text{photons} \longrightarrow \text{one bright band on screen}$$

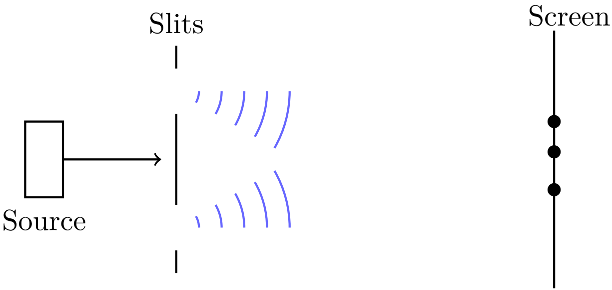
- Each photon travels in a straight line through the slit.
- Nothing strange here, fully classical behaviour.

Quantum Setup: One Photon at a Time

- Now imagine sending photons **one at a time**.
- No two photons overlap, no waves piling up.
- With one slit open, the pattern stays a single band.
- Everything appears classical. . . so far.

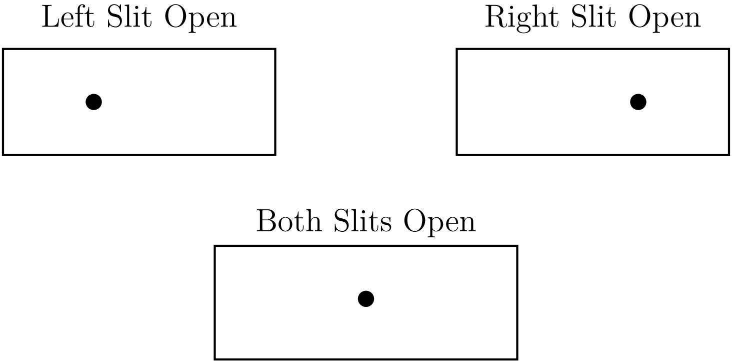
But when we open both slits, the very first photon behaves differently.

Double Slit Experiment



Two Slits: Something Impossible Happens

- The first photon lands in a different place depending on which slits are open.
- Even though each photon goes through only **one** slit.



With both slits open, the first photon lands somewhere new.

Why This Cannot Be Classical

- Classically, a particle should only care about:

$$\text{its path and the slit it goes through}$$

- In our diagrams:
  - The photon goes through **one** slit,
  - Yet its landing position changes if a **different slit** is opened.
- This cannot be explained by any classical trajectory.
- The only consistent picture:

$$\psi = \psi_1 + \psi_2$$

where each term carries a **complex phase**.

Interference Needs Phase

- In the double slit, the photon does not choose a slit.
- Its **wavefunction** explores both paths.
- These paths carry different **phases**.
- When phases add:

$$\psi = \psi_1 + \psi_2$$

  the result can be constructive or destructive.
- **Interference** is what shifts the landing spot.

Where Does Phase Come From?

- A classical wave has oscillations in time and space.
- Quantum mechanics copies this behaviour mathematically:

$$\psi \propto e^{i\theta}$$

- This expression is only possible because of the imaginary unit:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

- Complex phase is the reason two paths can interfere.

Without i: No Oscillations

- A real exponential:

$$e^{-kt} \quad \Rightarrow \quad \text{decay only}$$

- A complex exponential:

$$e^{-i\omega t} \quad \Rightarrow \quad \text{oscillation}$$

- Replace i with a real constant $\rightarrow$ oscillation disappears.
- Without oscillation, paths cannot develop different phases.
- Without different phases, there is **no interference**.

Without Interference: No Quantum Behaviour

- The double-slit pattern arises because:

$$\psi_1 + \psi_2 \quad \text{adds with phases}$$

- Remove the phase $\rightarrow$ both slits behave like one slit.
- No fringes, no shifting, no "spooky" behaviour.
- **Quantum mechanics collapses into classical diffusion.**

The Link Between the Weirdnesses

- Imaginary numbers feel “unreal” in mathematics.
- Quantum mechanics feels “unreal” in physics.
- But these two weirdnesses are actually the same thing:

$$\text{Quantum weirdness} \quad \Longleftrightarrow \quad \text{Complex phase from } i$$

- Without i, the universe would behave classically.

Quantum Tunnelling Needs Phase

- Classically, a particle cannot cross an energy barrier if

$$E < V_{\text{barrier}}$$

- Quantum mechanically, the wavefunction has an oscillatory phase:

$$\psi \propto e^{i\theta}$$

- This phase allows part of the wavefunction to “leak” through.
- **Tunnelling exists only because the wave evolves with a complex phase.**
- Without i, tunnelling disappears entirely.

Tunnelling and Stellar Fusion

- In the Sun, protons must fuse to form helium.
- Classically: the Coulomb barrier is too high, fusion should not happen.
- Quantum tunnelling allows protons to get close enough to fuse.
- Therefore:

$$\textbf{No imaginary unit } i \implies \text{no tunnelling}$$

$$\text{no tunnelling} \implies \text{no stars}$$

- The Sun shines because of a phenomenon that requires i.

Quantum Mechanics and the Early Universe

- The early universe was nearly symmetric:

$$\text{matter} \approx \text{antimatter}$$

- Tiny asymmetries allowed a small excess of matter to survive.
- These asymmetries come from processes with complex phases.
- Without those phases (i.e. without i):

$$\text{matter–antimatter} \to \text{perfectly symmetric}$$

- The universe would contain no matter at all.

A Number Born from Mathematics

- Mathematics grows by internal consistency.
- $\sqrt{-1}$ forced us to create a new number.
- i was originally seen as “unreal” or suspicious.
- It emerged from maths, not from physics.

But Physics Needed It

- Schrödinger's equation requires i.
- Without i:
  - no oscillations
  - no interference
  - no tunnelling
  - quantum mechanics collapses to diffusion
- Without quantum mechanics:
  - stars cannot shine
  - nuclear fusion cannot happen
  - early-universe fluctuations cannot form structure

We Exist Because the Universe Runs on i

- Quantum fluctuations seeded galaxies and stars.
- These fluctuations require complex phase.
- Without phase:

$$\text{no galaxies, no chemistry, no life}$$

- A number once called "imaginary" is fundamental to reality.

Imaginary numbers are part of the universe's architecture.

Questions

Thank you.